



Outsourcing, subcontracting-in and radical innovativeness

The moderating effect of manufacturing strategy

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Abstract

Purpose – The authors aim to develop and test hypotheses that link outsourcing and subcontracting-in activities of small high-tech firms to their radical innovativeness. In addition, they seek to investigate how a firm's strategy moderates the associations between their outsourcing and subcontracting-in activities and radical innovativeness.

Design/methodology/approach – The authors utilized regression analytical technique and categorical moderation analytical technique to test their hypotheses on survey data of 579 firms.

Findings – Results show that outsourcing has a positive association with radical innovativeness. In contrast, subcontracting-in shows a negative association with radical innovativeness. Finally, the influence of both outsourcing and subcontracting-in activities on radical innovativeness are contingent upon a firm's manufacturing strategy.

Research limitations/implications – There are potential limitations relating to the authors' use of secondary data. There is a need to investigate the processes through which outsourcing and subcontracting-in relate to innovation performance.

Practical implications – An implication of this study is that in order to develop radical innovativeness, firms need to consider their strategic or competitive inclination when evaluating their outsourcing and subcontracting-in decisions and activities.

Social implications – There are also social implications since outsourcing and subcontracting-in activities involve social relationships.

Originality/value – Linking boundary spanning activities of firms to innovation performance represents a contribution to the literature. Further, establishing that the effectiveness of such boundary activities depends on a firm's specific manufacturing strategy represents a contribution.

Keywords Manufacturing strategy, Innovation, Outsourcing, Subcontracting, SMEs

Paper type Research paper



1. Introduction

In the recent past, there has been an increased recognition of the influence of outsourcing and subcontracting-in on firm performance (De Koster and Balk, 2008; Gray *et al.*, 2009; Handley and Benton, 2009; Bichescu *et al.*, 2009). Outsourcing by a firm is defined as the delegation or farming out of peripheral tasks to others thereby enabling the firm to focus more on its core competencies (Mayer and Salomon, 2006). Subcontracting-in

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on the other hand is similar to contract manufacturing and is defined as a situation whereby a firm undertakes operational related activities on behalf of a client thereby enabling the firm to leverage its internal capabilities (Cachon and Harker, 2002).

While the importance of both outsourcing and subcontracting-in has been recognized, there has been a lack of consensus in the literature on their relationship with innovation performance. For example, it has been suggested that outsourcing and subcontracting-in allow for better leveraging of the skills and competences of the firms involved (Gallouj and Weinstein, 1997) which may lead to shorter product development times and reduced development costs (Wasti and Liker, 1999). On the other hand, it has been argued that by outsourcing, firms may risk eroding their own internal capabilities (Windrum *et al.*, 2009; Anderson and Parker, 2002) which may restrict the firm's long-term ability to innovate (i.e. innovativeness; Deshpande *et al.*, 1993).

A firm's innovative capability may depend on the type of knowledge that it draws upon or that is generated from outsourcing and subcontracting-in (Subramaniam and Youndt, 2005). How this type of knowledge that is gained from such boundary level activities of a firm is translated to internal innovative capabilities is yet to be addressed in the literature. Given the prevalence of outsourcing and subcontracting-in and the importance of innovativeness on a firm's long-term vitality (Deshpande *et al.*, 1993; Hult *et al.*, 2004b), there is a need to investigate the relationship between these key firms' sourcing strategies and innovativeness especially in the context of small firms that depend on their supply chain constituents to provide much of the peripheral activities related to their operations (Kaufmann and Todtling, 2002).

In addition, the conditions under which outsourcing and subcontracting-in are more or less likely to be effective in enhancing innovative capabilities of firms have not been investigated in previous studies. We identify manufacturing strategy (Youndt *et al.*, 1996; Klassen and Whybark, 1999) as a contingent variable in the relationship. A firm's manufacturing strategy is an indication of how it intends to achieve stated objectives in order to achieve its long-term goals (Miller and Roth, 1994; Bozarth and McDermott, 1998). Therefore, we would expect a firm's manufacturing strategy to influence the effectiveness of any major activities of a firm including outsourcing and subcontracting-in. However, while much research has been done in developing manufacturing strategy classifications and taxonomies and the effect that different manufacturing taxonomies have on financial performance (Amoako-Gyampah and Acquah, 2008; Vickery *et al.*, 1994; Wathen, 1995), we are not aware of any study that has investigated how different manufacturing strategies influence the relationship that outsourcing and subcontracting-in have on non-financial outcomes such as the innovativeness of a firm. Yet, such an investigation is important to the understanding of the totality of the effectiveness of outsourcing and subcontracting-in in developing innovative capabilities.

In sum, we had two main objectives in this study. First, we use the knowledge based view (KBV; Grant, 1996b; Nonaka, 1994) to motivate and test hypotheses linking outsourcing to radical innovativeness and subcontracting-in to radical innovativeness in small firms. A focus on small firms is merited, since they are more likely to draw knowledge from and benefit from outsourcing and subcontracting-in due to their limited resources. Second, we adopt Miller and Roth's (1994) manufacturing strategy taxonomy of caretakers, marketeers, and innovators as a framework to investigate the moderating role of manufacturing strategy on the relationship between outsourcing, subcontracting-in, and radical innovativeness in small firms. This study intersects the fields of innovation,

supply chain management and manufacturing strategy. It contributes to the extant literature by extending the KBV into these domains to address the blurring of firm production boundaries and knowledge flows within the supply chain. Specifically, we seek to understand the influence of outsourcing and subcontracting-in on a firm's capability to produce radical innovations and how a firm's choice of manufacturing strategy might enhance or inhibit these relationships.

In the next section, we discuss the theoretical framework for the study. We then discuss and define radical innovativeness. Next, we explain the influence that outsourcing and subcontracting-in have on radical innovativeness and develop the first set of study hypotheses. Then, we discuss the moderating role of manufacturing strategy. We present the methods and analytic strategy and then discuss the results and present the conclusion to the study.

2. Theoretical and conceptual background

In the strategy literature, the core competency paradigm (Prahalad and Hamel, 1990) has been tapped to explain outsourcing decisions of firms. The basic argument of core competency is that firms can leverage skills and resources by strategically focusing on core activities and strategically outsourcing non-core activities. The core competency paradigm is analogous to the focused factory concept (Skinner, 1974) in the operations management literature, which involves ensuring that a firm's manufacturing task is simple and focused on a narrow mix of core capabilities (Vokurka and Davis, 2000). By narrowing their responsibilities manufacturers enhance their learning through repetition, limit their "conflicting production tasks" (Skinner, 1974, p. 111) and, therefore, improve their overall performance.

These two views provide an insight into the motivation for outsourcing (i.e. to ultimately improve performance). However, they seem silent in explaining the influence that outsourcing and subcontracting-in may have on innovative capabilities. This, perhaps is because knowledge acquisition, learning and integration are not highlighted elements of the paradigms. In contrast, the KBV (Nonaka, 1994; Grant, 1996b) is a theory that justifies firms' existence based on their acquisition and use of knowledge. It is based on the tenet that the generation of innovative capabilities by firms depends on the type of knowledge gained and how such knowledge is used by the firm. As such, KBV can be used to provide an insight into how outsourcing and subcontracting-in may be related to firms' innovative capabilities.

KBV considers firms as heterogeneous bundles of knowledge assets (Grant, 1996b). While individuals are fundamental constituents in generating knowledge, they can only gather specialized knowledge. The role of the organization is to develop conditions so that the specialized knowledge of individuals can be integrated to develop new forms and, therefore, generate new products and services (Kogut and Zander, 1992). According to the KBV, organizations develop such conditions through two mechanisms: routines and directions (Grant, 1996b). Routines help to enhance learning through repeated performance. Directions help to identify which knowledge is more important to focus on. Using directions and routines, firms create new types of knowledge through reconfiguration of existing knowledge; which generates incremental knowledge patterns or through extension of their knowledge to generate radically new patterns (Grant and Baden-Fuller, 1995). KBV is applicable to this study because outsourcing and subcontracting-in may result in the gathering, integrating and extension of internal and

external knowledge (Hult *et al.*, 2004a) which according to KBV lead to the production of new innovations that may be incremental or radical in nature.

2.1 *Radical innovativeness*

Innovativeness has been defined as a firm's capability to develop and introduce new products, services or processes (Hult *et al.*, 2007; Lovelace *et al.*, 2001). Innovativeness requires the willingness and receptivity to take on new challenges (Garcia *et al.*, 2003). A firm's capability to produce radical innovations that are completely new to the world and can change the basis of competition or represent a fundamental change to existing related innovations, can be described as radical innovativeness (Subramaniam and Youndt, 2005). Underlying the ability to develop radical innovations is, therefore, the ability to conduct wide and deep searches in developing new and distant knowledge (Zhang *et al.*, 2009). Thus, radical innovativeness is the underlying capabilities of a firm to generate fundamentally different innovations through wider and deeper explorations. Outsourcing and subcontracting-in may involve some level of collaboration and partnership between participating firms. According to the KBV, this type of collaboration may lead to the transfer and reconfiguration of knowledge thereby developing the capability of participating firms to produce new materials that may or may not be radically innovative (i.e. radical innovativeness). Hence, the radical innovativeness of a firm is an appropriate outcome variable for capturing the effect of outsourcing and subcontracting-in.

Measures such as patents (Rosenkopf and Nerkar, 2001) which could potentially result in radical innovations, and which arguably capture a firm's ability to conduct deep and wide searches to develop distant knowledge can be used to measure a firm's radical innovativeness. Patents have been used as a reliable measure of radical innovation activity in previous studies (Acs and Audretsch, 1989; Amara *et al.*, 2008). Further, since patents can be linked to knowledge integration in firms (Rosenkopf and Nerkar, 2001), we posit that the level of a firm's patent activity would represent a suitable measure of radical innovativeness in this study.

2.2 *Outsourcing and radical innovativeness*

As we have previously defined, outsourcing in our study refers to the process by which a principal firm farms out all or part of its manufacturing operations to a subcontractor or contract manufacturer to produce an item based on a customized specification (Webster *et al.*, 1997). Outsourcing enables firms to focus their limited resources on their core businesses by allowing the "farming out" of standardized, routine and non-core tasks. Thus, a firm that outsources its non-core tasks could enhance its capability to develop new products and services (Li *et al.*, 2008) as a result of more focus on knowledge-based activities.

According to the KBV, the conditions that organizations create for the integration and application of individuals' knowledge to produce innovations is fundamental to its performance (Kogut and Zander, 1992). For example, a basic assumption of the KBV is that knowledge is a preeminent productive resource and that it is created and stored within individuals that make up an organization (Grant, 1997). As such, by outsourcing non-core activities and focusing on core competencies, an environment may be created whereby individuals' knowledge within a firm can be exploited, reconfigured and integrated to develop capabilities that can help with deeper and wider explorations. Such an

environment has been found to be vital for transforming knowledge and developing capabilities that can generate radical innovations (Quinn, 1999; Chesbrough and Teece, 1996). Further, the KBV notes the importance of direction in the integration of knowledge. By outsourcing, firms attain direction as they narrow their responsibilities and focus their resources on their core competencies. Direction helps firms to exploit knowledge gaps and opportunities resulting in capabilities relevant for developing radical innovation outcomes (Bogner and Bansal, 2007; Grant and Baden-Fuller, 2004). Therefore, on the basis of theory and literature we suggest the following hypothesis:

H1. Outsourcing is positively associated with radical innovativeness.

2.3 Subcontracting-in and radical innovativeness

Subcontracting-in involves undertaking the production of related operational activities on behalf of a client (Furlan *et al.*, 2007). Subcontracting-in generally requires the firm to restrict its focus to demands or needs that are specific to that of the client (Bichescu *et al.*, 2009). For example, build-to-print companies and capacity providers may act as subcontractors to other firms (Cagliano *et al.*, 2005). Thus, the subcontractor firm's resources and skills may be exclusively dedicated to the needs of client expectations (Webster *et al.*, 1997; Sousa and Voss, 2007). The implication of this is that subcontracting-in may restrict the firm's knowledge acquisition to that which is pertinent to the specific job that is being subcontracted-in. The narrowed focus on immediate customer needs may limit the firm's access to knowledge that is mainly related to improving service delivery to the customers such as the process or the efficiency aspects of the business. As a result, subcontractor firms may not have the opportunity to explore and exploit new knowledge that are transformational and can foster radical innovations. This is especially true for small firms because of their limited financial and human resources.

Further, from the perspective of KBV, subcontracting-in may lead a subcontractor to focus on knowledge that is already developed by the client. This suggests that by subcontracting-in, a firm will be primarily engaged in "reconfiguring existing knowledge" (Grant, 1996a, p. 385) to generate incremental outcomes as opposed to "encompassing new knowledge" (Grant, 1996a, p. 385) that can generate radical outcomes. Insights from the KBV would suggest that the knowledge gained by a subcontractor would be related to the reconfigurations of existing knowledge through routines as opposed to the development of new configurations (Bogner and Bansal, 2007; Grant and Baden-Fuller, 2004). The emphasis on routines that subcontracting-in encourages, may prevent the necessary focus required for directing the firm's human and capital resources towards developing capabilities for radical innovations. Therefore, we would expect subcontracting-in in a subcontractor firm to be negatively related to the capability to generate radical innovations resulting in the following hypothesis:

H2. Subcontracting-in is negatively associated with radical innovativeness.

3. Manufacturing strategy classifications

Manufacturing strategy is a functional sub-strategy that has been described as the task that a firm's operation must accomplish (Skinner, 1978) and the structural and infrastructural choices that it makes in order to accomplish such a task (Youndt *et al.*, 1996; Klassen and Whybark, 1999). Several studies have developed manufacturing strategy typologies and taxonomies. Among these, the most predominant is that of

Miller and Roth (1994) who leverage competitive priorities (i.e. cost, delivery, quality, and flexibility) to categorize firms into three groups of caretakers, marketeers and innovators. Caretakers distinguish themselves based on their focus on price, delivery dependability, speed, and conformance quality. Marketeers distinguish themselves based on their focus on market oriented competitive capabilities including offering broad product lines and volume flexibility. Innovators distinguish themselves based on their relative focus on design flexibility and new product flexibility.

Several contributions on the manufacturing strategies taxonomy have since followed Miller and Roth's seminal work. The focus of many of these studies (Kathuria, 2000; Lai *et al.*, 2007; Sum *et al.*, 2004) on developing manufacturing strategy taxonomies underscores the importance of such strategies in the competitive activities of firms. For a detailed literature review on manufacturing strategies taxonomies, please see Martin-Pena and Diaz-Garrido (2008) who conducted a thorough summary of the literature on the taxonomies of manufacturing strategies and the classification variables used to develop the strategies. In this study, we investigate how a firm's manufacturing strategy acts as a contingency factor that influences the effectiveness of outsourcing and subcontracting-in in the firm.

3.1 Moderating effects of manufacturing strategy on the link between outsourcing and radical innovativeness

In spite of the contributions to the development of manufacturing strategy taxonomies discussed above, literature suggests that the original taxonomy developed by Miller and Roth (1994) characterizes and captures the generality of firms in terms of the way that firms define and accomplish their manufacturing task (Skinner, 1978) and the taxonomy is thus adopted in this study. Indeed, there is enough support for Miller and Roth's (1994) strategy classification and it has been suggested that the distinct groups of caretakers, marketeers and innovators behave consistently differently in many contexts (Christiansen *et al.*, 2003; Frohlich and Dixon, 2001; Zhao *et al.*, 2006).

Clearly, a firm's focus on a specific competitive priority or manufacturing competence is likely to affect the way that the firm integrates or reconfigures knowledge during innovation or new product development (Yang, 2005). As we have argued previously, by outsourcing its non-core tasks, a firm may be able to focus on knowledge based activities by gleaning and integrating knowledge from within and from external constituents such as customers and suppliers. KBV would suggest that a marketeer firm is able to fully exploit the opportunity provided through outsourcing because of its focus on market oriented competitive capabilities (Miller and Roth, 1994) since such capabilities are important for effectively integrating knowledge and connecting the customer to the product for improved innovation performance.

Similarly, outsourcing can enable innovators to focus on developing their core competencies of introducing new designs and new products quickly. KBV would argue that both marketeers and innovators are able to leverage the additional internal resources generated through outsourcing to further develop and integrate knowledge that is specialized in line with their particular competitive capabilities. As Grant (1996a, p. 385) notes, this additional development of internal resources can increase "the breadth of specialized knowledge the organizational capability draws upon" resulting in an enhanced capability to develop radical innovations. Compared to caretakers, innovators share characteristics relating to product performance with marketeers (Miller and Roth, 1994).

Thus, although they represent distinct manufacturing strategies, in terms of the capability to develop radical innovative outcomes, we would expect the influence of marketeers and innovators on the effectiveness of outsourcing to be similar resulting in the following hypothesis:

- H3.* The positive relationship between outsourcing and radical innovativeness will not be significantly different between marketeers and innovators.

However, when compared with caretakers we would expect both marketeers and innovators to influence the relationship between outsourcing and radical innovativeness differently. According to KBV outsourcing allows a firm to have the direction and environment to focus on knowledge-based activities which can be exploited to develop radical innovative capabilities. As we have argued, because of their focus on market oriented and product performance capabilities, marketeers and innovators, respectively, are better able to exploit the opportunity to develop innovative capabilities through outsourcing. As such, we would expect the influence of outsourcing on radical innovativeness to be higher for marketeers than for caretakers. Similarly, we would expect the influence of outsourcing on radical innovativeness to be higher for innovators than for caretakers since outsourcing enables the former to focus on design and new product performance that can lead to radical outcomes compared to the caretakers' low emphasis on those competitive capabilities. In other words, a firm whose main priority is to provide high quality innovative products, may gain more benefits by outsourcing its routine activities to others to allow for its internal resources to be more focused on the development of radical innovative capabilities. In contrast, caretakers' low emphasis on innovative competitive capabilities implies that their ability to leverage the additional resources made available through outsourcing to develop radical innovative capabilities would be limited. Thus, our argument above would suggest that manufacturing strategies that are specific to both innovators and marketeers would influence the relationship between outsourcing and radical innovativeness more than would strategies specific to caretaker firms. Therefore, we offer the following hypotheses:

- H4.* The positive relationship between outsourcing and radical innovativeness is stronger (a) for marketeer firms than for caretaker firms and (b) for innovator firms than for caretaker firms.

3.2 Moderating effect of manufacturing strategy on the link between subcontracting-in and radical innovativeness

We have argued previously that subcontracting-in is negatively related to radical innovativeness because the subcontractor firm may focus on jobs that lead to the augmentation of existing skills. Further, by subcontracting-in a firm may be able to integrate and reconfigure existing knowledge resulting in the development of capabilities that may lead to incremental rather than radical innovative outcomes (Grant, 1996a). Since the focus of the firm is on the subcontract work that it does for a client, its efforts and resources will be focused on this activity irrespective of its own competitive inclination.

Because of their focus on price, delivery dependability and speed, caretakers' priorities are more aligned with the demands of subcontracting-in compared with priorities of marketeers and innovators. However, such performance may have little or no bearing on radical innovativeness. This is because, according to KBV, subcontracting-in may lead

to the reconfigurations of existing knowledge through routines. The focus on routines in subcontracting-in works against the development of radical innovativeness. As such, the performance of a caretaker subcontractor firm may be more aligned with the enhancement of efficiencies and development of incremental improvements rather than radical innovations. In contrast, the market oriented strategies and the design focus of both marketeers and innovators, respectively, may encourage radical innovativeness even within a subcontracting-in environment. This implies that the influence of both marketeers and innovators may decrease the negative influence that subcontracting-in have on radical innovativeness. In other words, although subcontracting-in activities tend to divert a firm's focus away from radical innovativeness, the fact that marketeers are inherently more market-orientated than caretakers means that marketeers may be in a constant drive to develop radical innovative capabilities and this may decrease the negative relationship that subcontracting-in activities have on radical innovativeness for marketeers compared to caretaker firms. Similarly, because the focus of innovators is inherently more aligned towards radical innovative outcomes than the focus of caretakers, innovators' focus may lead to a decrease in the negative influence of subcontracting-in on radical innovativeness for innovators compared to caretakers. Therefore, in relative terms, we would expect the negative influence of subcontracting-in activities on radical innovativeness to be stronger in caretaker firms than in both marketeers and innovator firms. Hypothesis 5 captures this notion below, followed by a conceptual model shown in Figure 1:

H5. The negative relationship between subcontracting-in and radical innovativeness is (a) stronger for caretaker firms than for marketeer firms and (b) stronger for caretaker firms than innovator firms.

4. Methods and data

4.1 Data set and measures

We use data collected by the Centre for Business Research at University of Cambridge and supplied by Council of European Social Science Data Archives (CESSDA). The objective of the survey, which was collected between the year 2000 and 2002 was

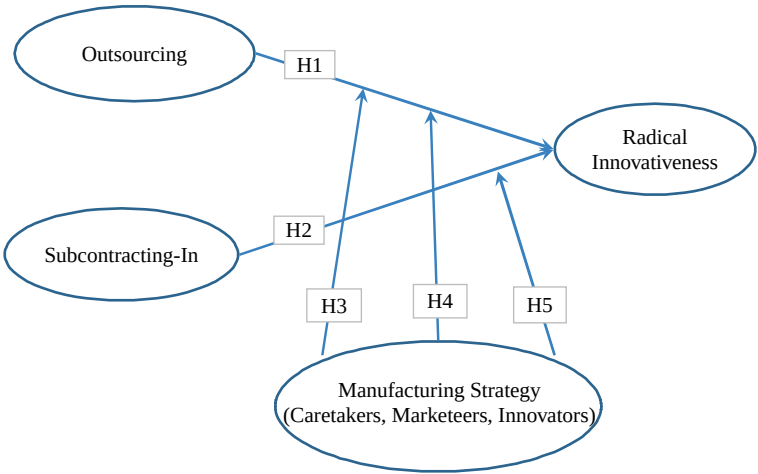


Figure 1.
Conceptual framework

to identify innovative behaviors in small firms operating in high-tech industries. A total of 579 cases with a combined response rate of 13.3 per cent were collected as part of the study. Although the response rate is relatively low, it is consistent with similar studies in the operations management literature (Song and Benedetto, 2008; Rosenzweig *et al.*, 2003; Swink *et al.*, 2006). Median age of the firms in the sample was 24 and median size of the firm was 41 employees. Survey respondents included those in ten major high tech industrial segments (based on UK's Standard Industrial Code, primary digit index). The largest five groups included those in pharmaceuticals manufacturing, industrial process control equipment, electronic valves, navigation instruments, software, and telecommunications services. The largest single group constituted 17 per cent of the sample size.

We measured radical innovativeness using the count data of the number of patent applications with respect to radical products, services and processes developed by the business (Ostergard, 2000; Amara *et al.*, 2008). Although the number of patent applications does not exactly reflect radical innovation of the firms, it is indicative of the extent to which firms actively pursue innovation. This measure has been used and justified in previous studies (Debackere *et al.*, 1999; Popp, 2005; Greenhalgh and Rogers, 2006; Tseng and Wu, 2007; Watanabe *et al.*, 2001). The extent of subcontracting-in was measured as work carried out on a subcontract basis for other firms as percent of total sales. Similarly, the extent of outsourcing was measured as work undertaken by the other firms for the business on a subcontract basis as a percent of total sales. Manufacturing strategy was measured using 11 questions that asked about different competitive capabilities of the firm. The questions asked for the importance of each competitive capability using a five point scale. For example, it asked how the respondents rank technological or scientific expertise or specialized product or service as their main competitive advantage. We present a discussion of the steps taken using multiple discriminant analysis (MDA) for developing manufacturing strategy variables in the upcoming sections.

4.2 Control variables

We controlled for organizational size, industry sector, country, company age, R&D intensity and technical expertise. Organizational size is frequently considered as an influence on firm level innovation and performance in small firms' research (Kimura, 2002). We controlled for organizational size using logarithm of yearly sales figures. Second, we controlled for industry sector since the speed of progress and emphasis on innovation can be different for different industries (Carillo, 2005). The data included firms from ten industrial groupings, namely chemicals, instruments, computers and electronics, machinery (e.g. aerospace), other manufacturing, telecommunication services, computer services, R&D services and technical testing, architectural and engineering services. We captured differences in emphasis on innovation, using the percent R&D expenditure for the year 2000 using Organization for Economic Cooperation and Development (OECD) Stat Extracts (labeled as industrial group in the results tables). We controlled for company age, since more established firms may have the advantage of time to develop innovative capabilities (Salimath *et al.*, 2008). We controlled for country since the data set included firms from the UK and Japan. R&D intensity, measured based on total expenditure in R&D activities as percent of sales, was included since the level of R&D intensity can influence the innovativeness of the firm.

Technical expertise was measured by based on the percent of full time employees that had professional and academic qualifications and certifications.

Organizational size, country and company age are prevalently used control variables since they can influence firm performance in general and innovation capabilities in particular. Beyond these three, our inclusion of industry group (based on level of R&D intensity), technical expertise (percent of professionals) and firm level R&D focus (R&D expenditure as percent of sales) was to capture possible endogeneity (i.e. selection bias) effects (Shaver, 1998). The concern arises when independent variable(s) could be correlated with the error terms of dependent variable. In our case, it is likely that both subcontracting-in and outsourcing, and innovation are driven by common antecedent(s) leading to possible endogeneity issue. The latter three control variables help segregate this issue.

4.3 MDA and strategy categorizations

The data set used included 11 measures of manufacturing capability dimensions (Table I). However, given that we used an archival data set for the study, the wording of the measures did not precisely match what Miller and Roth used. In order to accurately categorize the measures into the Miller and Roth's strategy categories, we requested for a panel of academics to develop five different combinations of how the 11 dimensions from the data set would fit the three strategies suggested. We used MDA to compare the five models; consistent with other studies that developed similar categories or taxonomies (Flynn and Flynn, 1999; Huang *et al.*, 2001).

MDA uses a linear combination of independent variables to identify differences between group means (Lachenbruch, 1975). To assess a three group categorization, MDA requires two functions to be generated (Hair *et al.*, 2005). We first verified the significance of the five suggested discriminant models based on the results of Wilk's lambda test for each (Klecka, 1980). All the five models showed significance for their two functions, suggesting that they are all acceptably discriminating (Table II). Goodness of fit for a MDA is increased as the canonical correlation (R^*) is raised and as the ratio of the eigenvalues for the two functions is more balanced (Rencher, 1992). Based on these results, DA-Model 5 was chosen as the most fitting. Table I shows the results of standardized canonical discriminant function coefficients for DA-Model 5, the selected model. Table II shows how the five DA models compare to one another.

Dimensions	Function 1	Function 2
Technological and scientific expertise	0.286	-0.263
Price/cost advantages	-0.376	0.732
Marketing and promotion	-0.110	0.610
Speed of service	-0.555	-0.115
Established reputation	0.007	-0.046
Design of product/service	0.505	-0.300
Quality of product/service	0.036	0.305
Specialized product/service	0.122	-0.167
Range of product/service	0.047	0.535
Personal attention and responsiveness to client needs	-0.214	0.398
Being first in the market with new products/service	0.460	-0.178

Table I.
Standardized canonical
discriminant function
coefficients for selected
model (DA-Model 5)

Notes: Significant at: $*p < 0.05$, $**p < 0.01$ and $***p < 0.001$; $n = 579$

	DA-Model 1			DA-Model 2			DA-Model 3			DA-Model 4			DA-Model 5 ^a		
	EV	R [*]		EV	R [*]		EV	R [*]		EV	R [*]		EV	R [*]	
Function 1	0.913	0.691		1.046	0.715		0.796	0.666		0.503	0.578		0.844	0.676	
Function 2	0.198	0.406		0.335	0.501		0.645	0.626		0.084	0.279		0.513	0.641	
	Wilks's λ			Wilks's λ			Wilks's λ			Wilks's λ			Wilks's λ		
Function 1	0.437 ^{***}	395.4		0.366 ^{***}	475.2		0.339 ^{***}	505.8		0.614 ^{***}	242.1		0.384 ^{***}	432.8	
Function 2	0.835 ^{***}	86.0		0.749 ^{***}	136.6		0.608 ^{***}	232.4		0.922 ^{***}	40.05		0.708 ^{***}	156.3	

Notes: Significant at: ^{*} $p < 0.05$, ^{**} $p < 0.01$ and ^{***} $p < 0.001$; ^aModel 5 is the best fitting model based on highest and closer eigenvalues and highest Wilks's λ values for the two functions; R^{*} – canonical correlation; EV – eigenvalue

Table II.
Discriminant analysis
comparisons

Using DA-Model 5, we developed a score for how each firm fits into the three strategic groups. In other words, each firm received a score as a caretaker, marketer and an innovator by averaging the score of all items that were categorized for the particular strategic group. Based on the highest strategic score, firms were categorized as predominantly belonging to one of the three strategic groups. For example, a firm with a marketer score of 7, a caretaker score of 6.5 and an innovator score of 7.5 would be categorized as an innovator. Firms with a tie in their highest scores were excluded from the study. A total of 56 firms had a tie in their highest strategy scores. This left us with 523 firms categorized into one of the three strategic groups of which 193 were predominantly caretakers, 133 were predominantly marketers and 197 were predominantly innovators. The means and standard deviation of the three manufacturing strategic groups on the set of 11 competitive capability variables are described in Table III. For each of the 11 competitive capabilities, the probability that one or more of the strategic group means

Competitive capabilities	Manufacturing strategic group			<i>F</i> = value (<i>p</i> = probability)
	Group 1 caretakers	Group 2 marketeers	Group 3 innovators	
<i>Technological and scientific expertise</i>				
Cluster mean	3.76	3.76	4.37	<i>F</i> = 17.33
SE	1.02	0.962	0.849	<i>p</i> = 0.000
<i>Price/cost advantages</i>				
Cluster mean	3.94	3.00	3.01	<i>F</i> = 48.07
SE	0.801	0.872	0.975	<i>p</i> = 0.000
<i>Marketing and promotion</i>				
Cluster mean	2.87	3.50	2.91	<i>F</i> = 10.34
SE	1.12	0.982	1.05	<i>p</i> = 0.000
<i>Speed of service</i>				
Cluster mean	4.11	4.05	3.52	<i>F</i> = 17.52
SE	0.797	0.886	0.893	<i>p</i> = 0.000
<i>Established reputation</i>				
Cluster mean	3.54	3.84	3.72	<i>F</i> = 2.34
SE	1.16	1.04	1.07	<i>p</i> = 0.072
<i>Design of product/service</i>				
Cluster mean	2.90	3.19	3.86	<i>F</i> = 26.45
SE	1.10	1.07	0.944	<i>p</i> = 0.000
<i>Quality of product/service</i>				
Cluster mean	3.92	4.23	4.23	<i>F</i> = 6.06
SE	0.900	0.858	0.739	<i>p</i> = 0.000
<i>Specialized product/service</i>				
Cluster mean	3.58	3.86	4.23	<i>F</i> = 15.0
SE	1.03	0.990	0.849	<i>p</i> = 0.000
<i>Range of product/service</i>				
Cluster mean	3.13	3.64	3.40	<i>F</i> = 6.48
SE	1.07	0.969	1.02	<i>p</i> = 0.000
<i>Attention and responsiveness</i>				
Cluster mean	4.19	4.42	4.08	<i>F</i> = 4.74
SE	0.876	0.686	0.860	<i>p</i> = 0.003
<i>Being first in the market</i>				
Cluster mean	2.77	3.05	3.92	<i>F</i> = 32.19
SE	1.22	1.30	1.26	<i>p</i> = 0.000

Table III.
Competitive capabilities
by group

is significantly different from another is included as well. These results show that the three strategic groups differed on ten of 11 variables at the 0.001 level of significance or less:

$$\text{RINOV} = a + b_1\text{OutS} + b_2\text{MS}_1 + b_3\text{MS}_2 + b_4\text{OutS}*\text{MS}_1 + b_5\text{OutS}*\text{MS}_2 + e \quad (1)$$

$$\text{RINOV} = a + b_1\text{SubIn} + b_2\text{MS}_1 + b_3\text{MS}_2 + b_4\text{SubIn}*\text{MS}_1 + b_5\text{SubIn}*\text{MS}_2 + e \quad (2)$$

In equation (1) outsourcing is represented by OutS, MS₁ is the first code variable for the moderator variable (manufacturing strategy) and MS₂ is the second code variable for the moderator variable. MS₁ relates the focal group (caretakers in this study) with a first compared group (marketeers). MS₂ relates the focal group with a second compared group (innovators). OutS*MS₁ and OutS*MS₂ represent the interaction terms between outsourcing and the manufacturing strategy moderating variable. Coefficients b₂ and b₃ represent the mean difference across groups. Coefficients b₄ and b₅ represent slope differences across groups. Equation (2) is used to test *H5a* and *H5b* in a similar fashion. We note here that for *H3*, we compared two groups (innovators and marketeers). This comparison requires only one code variable. As such, R-Models 5 and 6 in Table VI shows one strategy factor included (MSMI).

To adjust for differences in sample size among our three manufacturing strategy groups, we applied weighted effect coding (WEC). In WEC the assigned value for Z₁ and Z₂ are based on the sample sizes of the groups (Aguinis, 2004). For example, considering the caretaker group as a baseline for comparison (as in *H4*), the value for MS₁ is $-\frac{n_{\text{marketeer}}}{n_{\text{caretaker}}} (-133/193 = -0.70)$ and the value for MS₂ is $-\frac{n_{\text{innovator}}}{n_{\text{caretaker}}} (-197/193 = -1.03)$.

4.4 Direct effect and categorical moderation analysis

To test the direct effects of outsourcing and subcontracting-in on radical innovativeness, we conducted a series of negative binomial regression analyses. It will be recalled that our dependent variable (radical innovativeness) was based on the number of patents, and, therefore, not normally distributed. As such, consistent with previous studies (Miller *et al.*, 2007) we used negative binomial distributions rather than ordinary least squares regression to accommodate the “count data” distribution.

H3 and *H4* in our model required comparing the effects of different manufacturing strategies on the relationship between outsourcing and radical innovativeness. *H3* required comparing marketeers with innovators, whereas *H4* required comparing marketeers with caretakers and comparing innovators with caretakers. We applied a categorical moderation analytical technique (Aguinis, 2004). We chose this approach since our moderating variable (manufacturing strategy groups) comprised of two (for *H3*) and three different categories (for *H4*). In cases where a moderator variable comprises of three categorical groups, two code variables (i.e. MS₁ and MS₂) are necessary (Aguinis, 2004). In addition, the moderated regression model needs to include the two code variables and the quantitative predictor to fully represent the first order and the moderating effects (West *et al.*, 1996). The modeled equation (1) for radical innovativeness (displayed above) was used to analyze and test *H4*.

H5 required assessing the relative effects of subcontracting-in on radical innovativeness of the manufacturing strategies using a similar approach to *H4*. The modeled equation (2) was used to test *H5*.

5. Results

We apply a step-wise regression approach to test our hypotheses. Table IV shows the zero order correlation of the variables used. Table V shows the results for testing the direct effects of outsourcing and subcontracting-in on radical innovativeness. R-Model 4, which includes subcontracting-in and outsourcing along with the effects of our six control variables on radical innovativeness, was used to assess *H1* and *H2*. *H1* posited that outsourcing is positively associated with radical innovativeness. R-Model 4 in Table V shows the coefficient for OutS term to be positive and statistically significant (1.15; $p < 0.001$). Typical means for comparing regression models with count data is through Akaike information criterion (AIC) and Bayesian information criterion (BIC) (Gschlossl and Czado, 2008) which indicate that R-Model 4 shows a better fit compared to R-Models 1, 2 and 3. Based on the significant results, and the fact that R-Model 4 is a more fitting model, *H1* is supported. *H2* posited that subcontracting-in is negatively associated with radical innovativeness. R-Model 4 in Table V shows the SubIn term to have a negative and statistically significant coefficient (-1.32 ; $p < 0.001$). Thus, *H2* is supported.

To test the hypotheses regarding the moderating effect of the manufacturing strategy groups, we constructed two sets of stepwise regression models. The first set provided the test for *H3* (that the moderating influence of marketeers and innovators on the associations of outsourcing with radical innovativeness is not significantly different). R-Model 5 (Table VI) was developed by adding MSMI (the single categorical variable that allowed for comparing marketeers and innovators) to R-Model 4. We then added the interaction term OutS*MSMI to develop R-Model 6. The interaction term OutS*MSMI is not significant, supporting *H3*. To test our hypotheses comparing the differences in the moderating effects between caretakers and the other two groups (*H4a*, *H4b*), we first developed R-Model 7 (Table VI) by adding the two moderating factors (MS1 and MS2) to R-Model 2. We then added the interaction terms (OutS*MS1 and OutS*MS2) to develop R-Model 8. The interaction term OutS*MS1 is significant ($p < 0.001$) as is the interaction term OutS*MS2 ($p < 0.05$) in R-Model 10 suggesting that the influence of both marketeers and innovators on the relationship between outsourcing and radical innovativeness differs from caretakers' influence on the relationship. Therefore, *H4a* and *H4b* are both supported.

To test the effects of *H5* (on the difference in the moderating influence of caretakers versus marketeers and innovators on the relationship between subcontracting-in and radical innovativeness), we first developed R-Model 9 (Table VII) by including the control variables of the study. We then subsequently added SubIn to develop R-Model 10, the two moderating factors (MS1 and MS2) to develop model 11 and finally the interaction terms (Subin*MS1 and Subin *MS2) to develop R-Model 12. The interaction term Subin*MS1 is significant ($p < 0.05$) as is the interaction term Subin*MS2 ($p < 0.05$) in R-Model 12 suggesting that the influence of both marketeers and innovators on the relationship between outsourcing and radical innovativeness differs from caretakers' influence on the relationship. Therefore, *H5a* and *H5b* are both supported. Interpreting results of interactions solely by relying magnitude and directionality of interaction coefficients in general, and for categorical moderators in particular, may lead to misleading explanations (Aguinis and Stone-Romero, 1997). Therefore, we validated the findings by reviewing the graphical depictions of the relationships (Figures 2 and 3; Hoetker, 2007).

Variable	Mean	SD	1	2	3	4	5	6	7	8	9
1. Size (log sales ^a , control)	7.84	2.21	1								
2. Country (control)	1.59	0.492	0.269**	1							
3. Industry group ^b (control)	6.58	3.26	-0.085**	-0.065	1						
4. Age (years, control)	29.1	26.4	0.156**	0.198*	-0.053	1					
5. R&D intensity (% , control)	7.52	14.78	-0.468**	-0.138**	0.397**	-0.100**	1				
6. Technical expertise (control)	35.1	41.4	-0.117**	-0.237**	0.126*	-0.1146**	0.217**	1			
7. Subcontracting-in (%)	27.7	35.4	0.026	0.173*	-0.026	-0.018	-0.076	-0.057	1		
8. Outsourcing (%)	18.4	21.6	0.086	0.249*	-0.081	0.036	0.021	-0.048	0.099*	1	
9. Radical innovativeness	3.85	13.5	0.118**	0.162*	-0.019	0.091*	0.060	-0.023	-0.105*	0.189**	1

Notes: Significant at: * $p < 0.05$ and ** $p < 0.01$ (two-tailed); $n = 579$, ^ain million pounds sterling; ^bbased on percent R&D expenditure in each industrial sector as noted in OECD Stat Extracts

Table IV.
Means, standard
deviations, correlations

Table V.
Direct effects of
outsourcing and
subcontracting-in on
radical innovativeness
(H1 and H2)

Variables	R-Model 1	R-Model 2	R-Model 3	R-Model 4
Intercept	− 3.54 ***	− 3.45 ***	− 3.51 ***	− 3.40 ***
Size (log sales)	0.225 **	0.218 **	0.213 **	0.204 ***
Country	1.29 **	1.06 **	1.46 **	1.24 **
Age	0.009 ***	0.010 ***	0.009 ***	0.011 ***
Industrial group	3.22	3.321	3.244	3.43
R&D intensity (%)	0.036 ***	0.036 ***	0.034 **	0.034 **
Technical expertise			0.004	0.003
Outsourcing (Out S)		1.20 ***		1.15 ***
Subcontracting-in (SubIn)			− 1.36 ***	− 1.32 ***
Log likelihood	− 933.3	− 909.4	− 917.6	− 835.1
AIC	1,877	1,830	1,847	1,723
BIC (sample size adjusted)	1,896	1,854	1,870	1,688

Notes: Significant at: * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$; $n = 579$; coefficients are in log-counts

Table VI.
Moderating effects of
manufacturing strategy
(caretakers, marketeers
and innovators) on the
relationships between
outsourcing and radical
innovativeness
(H3 and H4)

Variables	R-Model 5 <i>H3</i>	R-Model 6	R-Model 7 <i>H4</i>	R-Model 8
Intercept	− 3.05 ***	− 3.08 ***	− 3.12 ***	− 3.08 ***
Size (log sales)	0.183 ***	0.185 ***	0.189 **	0.192 **
Country	1.17 **	1.16 **	1.023 **	0.985 **
Age	0.006 *	0.007 *	0.008 ***	0.010 ***
Industrial group	3.30	3.04	2.96	2.61
R&D intensity (%)	0.023 ***	0.023 ***	0.024 ***	0.023 ***
Technical expertise	0.003	0.002	0.002	0.001
Outsourcing (OutS)	1.33 ***	1.35 ***	1.38 ***	1.20 ***
MSMI	0.242 **	0.350 **		
OutS*MSMI		− 0.417 (ns)		
Mfg strategy factor 1 (MS1)			− 0.314 ***	− 0.664 ***
Mfg strategy factor 2 (MS2)			0.358 ***	0.405 ***
OutS*MS1				1.41 ***
OutS*MS2				0.510 *
Log likelihood	− 731.3	− 728.9	− 876.0	− 872.9
AIC	1,488	1,479	1,770	1,768
BIC (sample size adjusted)	1,526	1,519	1,813	1,807

Notes: Significant at: * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$; total $n = 23$, n for caretakers = 193, n for marketeers = 133, n for innovators = 197; caretakers are compared to marketeers (MS1) and innovators (MS2); ns – not significant

6. Discussions, contributions and conclusions

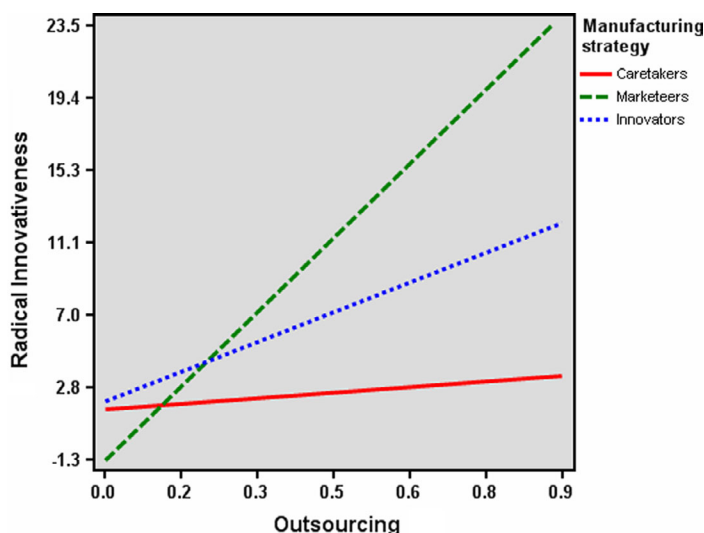
Radical innovativeness, or the capability to produce radical innovations (Subramaniam and Youndt, 2005) may be desirable for firms because such innovations may serve as the gateway to success in today's competitive environment (Dutta and Weiss, 1997). Thus, how outsourcing and subcontracting-in relate to radical innovativeness is of importance to practicing managers and to bridging the gap between the innovation and the supply chain management fields. We employed the KBV to investigate this relationship and

Table VII.

Moderating effects of manufacturing strategy (caretakers, marketeers and innovators) on the relationships between subcontracting-in and radical innovativeness (H5)

Variables	R-Model 9	R-Model 10	R-Model 11	R-Model 12
Intercept	-3.09 ***	-3.05 ***	-3.05 ***	-3.06 **
Size (log sales)	0.194 ***	0.183 ***	0.174 **	0.174 **
Country	1.23 **	1.41 **	1.44 ***	1.44 **
Age	0.006 *	0.005	0.005	0.006 *
Industrial group	3.15	3.36	3.08	3.16
R&D intensity (%)	0.026 ***	0.024 ***	0.021 ***	0.021 **
Technical expertise	0.002	0.003	0.003	0.002
Subcontracting-in (SubIn)		-1.35 ***	-1.24 ***	-1.22 **
Mfg strategy factor 1 (MS1)			-0.244 *	-0.127
Mfg strategy factor 2 (MS2)			0.273 **	0.292 **
SubIn*MS1				-0.419 *
SubIn*MS2				-0.073 *
Log likelihood	-784.8	-765.4	-760.8	-758.7
AIC	1,583	1,547	1,543	1,542
BIC (sample size adjusted)	1,617	1,589	1,589	1,587

Notes: Significant at: * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$; total $n = 523$, n for caretakers = 193, n for marketeers = 133, n for innovators = 197; caretakers are compared to marketeers (MS1) and innovators (MS2); ns – not significant



Notes: No significant difference between the influence of marketeers and innovators (H3 supported); marketeers influence is significantly more than caretakers influence (H4a supported); innovators influence is significantly more than caretakers influence (H4b supported)

Figure 2. Moderating effect of manufacturing strategy on the relationship between outsourcing and radical innovativeness

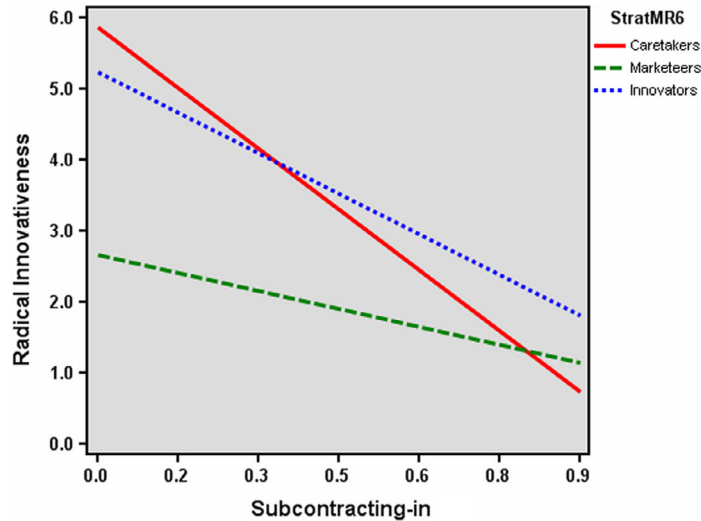


Figure 3.
Moderating effect of
manufacturing strategy on
the relationship between
subcontracting-in and
radical innovativeness

Notes: Caretakers' influence is significantly more negative than marketeers (*H5a* supported); caretakers' influence is significantly more negative than innovators (*H5b* supported)

found outsourcing to be positively associated with radical innovativeness. This finding is consistent with KBV suggestion that by outsourcing and focusing on core competencies, firms are better able to operate within an environment that allows for and fosters the development, exploitation, reconfiguration and integration of new knowledge to develop radical innovative capabilities. Further, we found subcontracting-in to be negatively related to radical innovativeness. Subcontracting-in relates to performing specific operational jobs. Working with clients generally involves sharing knowledge that is specific to the particular task at hand. According to KBV, this type of collaboration may lead to the reconfiguration of existing knowledge that may lead to the development of incremental innovative capabilities rather than radical innovative capabilities (Subramaniam and Youndt, 2005).

These linkages between outsourcing and radical innovativeness and between subcontracting-in and radical innovativeness contribute to knowledge since the findings extend the KBV into the supply chain management field and builds on the extant literature on innovation in supply chains that has typically focused on the involvement of suppliers in new product development. Further, research is needed to address the blurring of firm production boundaries and knowledge flows within the supply chain. Our study partly addresses this gap in the literature by shedding light on how boundary level firm activities can enhance internal innovative capabilities. In terms of implications, our results suggest that firms involved in outsourcing and subcontracting-in activities must strive to exploit the opportunities that such activities provide. Specifically, when activities are outsourced, managers can concentrate on harnessing and developing the radical innovative capabilities of the firm. Similarly when activities are contracted-in, managers must focus on how to exploit the opportunities that collaborating with clients provide. In particular, because contracting-in operational jobs tends to focus attention on improvements related

to the contracted jobs rather than on radical innovations, managers must take conscious steps towards achieving radical innovations.

Perhaps, the most appealing result of this study is the finding that the effects of outsourcing and subcontracting-in on radical innovativeness are contingent upon a firm's manufacturing strategy (i.e. marketeers, innovators and caretakers) as defined by Miller and Roth (1994). In other words, the strategies of manufacturing firms as defined by their competitive capabilities may play an important role in the effectiveness of their supply and purchasing strategies (Rosenzweig *et al.*, 2003). Such applications of manufacturing strategies as conditions under which outsourcing and subcontracting-in will be more or less effective in terms of enhancing innovativeness underscore the importance of the strategies and represent a contribution to the literature.

Specifically, our results reveal that in spite of their distinct competitive foci, there is no significant difference between the influence of marketeers and innovators on the relationship between outsourcing and radical innovativeness. Indeed, these results suggest that, from a manufacturing strategy perspective, there is an equifinality in the relationship between outsourcing and radical innovativeness for certain manufacturing strategies. Both marketeers and innovators can be equally radically innovative through an outsourcing strategy. Further, our results reveal that the effect of outsourcing on radical innovativeness is stronger for both marketeers and innovator firms than for caretakers indicating that the effectiveness of firms engaged in outsourcing to generate radical innovative capabilities is strongly tied to their competitive capabilities. Because of their focus on market oriented and product performance capabilities, both marketeers and innovators are better able to exploit opportunities obtained through outsourcing activities to nurture the capability to generate radical innovation outcomes than caretakers who are typically more internally focused on speed, price and delivery dependability. Both marketeers and innovators have distinct capabilities that can be leveraged when the extent of internal operational tasks are reduced (i.e. in outsourcing). In contrast, in spite of the opportunities that they may have through outsourcing non-core tasks, caretakers may not have the impetus to develop the capability for radical innovation outcomes because of their "low relative emphasis on the development of such competitive capabilities" (Miller and Roth, 1994, p. 290). The overall implication of these findings is that in order to develop radical innovativeness, firms need to consider their strategic or competitive inclination when evaluating their outsourcing and subcontracting-in decisions and activities. Specifically, in terms of achieving radical innovativeness, it does not appear to matter whether an outsourcing firm adopts a marketeer or an innovator strategy as both strategies tend to positively influence the effect of outsourcing on radical innovativeness. What matters is how the firm exploits and harnesses the opportunities provided through its outsourcing activities to achieve radical innovativeness. Our result that indicates a stronger influence of marketeer and innovator strategies on the link between outsourcing and radical innovativeness compared with the influence of caretaker strategy has some implications. Outsourcing firms that intend to achieve radical innovativeness must discourage caretaker strategy and adopt innovator and/or marketeer strategies as these strategies enable them to effectively harness and exploit opportunities provided through outsourcing to achieve radical innovativeness better than caretaker strategies.

Finally, our analyses reveal that the negative relationship between subcontracting-in and radical innovativeness is enhanced in caretaker firms compared to marketeers and innovator firms. According to KBV, the fairly repetitive and routine jobs done by

subcontractors lead to the reconfiguration of existing knowledge that may lead to incremental outcomes but negate the development of radical innovativeness. Our finding suggests that caretaker firms' focus on speed and delivery dependability helps to accentuate these relationships since these capabilities are more aligned with developing incremental outcomes. In contrast, the negative effect of subcontracting-in on radical innovativeness is decreased in marketeers and innovators. The implication of these findings is that although subcontracting-in discourages radical innovativeness in general, firms that engage in subcontracting-in would find it easier to achieve radical innovativeness under a marketeer or an innovator strategy compared to a caretaker strategy.

7. Limitation and future research

Our research should be evaluated within the context of its limitations. While there has been an increased use of secondary data in empirical research in the operations and supply chain management field (Roth, 2007), the limitations relating to our use of secondary data is worth noting. Our data did not provide specifics on the items or services that were subcontracted-in or outsourced. As such, we were unable to identify the "strategic" or "non-criticality" (Kraljic, 1983) of the items or services included in sourcing practices. However, because our sample firms was comprised of firms in manufacturing industries, we defined outsourced and subcontracting-in jobs in the sampled firms to be related to mainly operational activities. Moreover, we believe that our model is able to test this effect of the sourcing strategies on innovativeness on average. The issue of validity relating to whether the variables used may in fact be measuring something other than intended is worthy of note. Our use of a panel of academics to validate some of the constructs used in this study helps to alleviate the issues noted above. Also, we investigated research questions that have constructs with mainly objective measures in the data set. Thus, potential reliability and validity issues are minimized.

In this study, we investigated the influence of boundary level strategies such as outsourcing and subcontracting-in on radical innovativeness and the role of a firm's manufacturing strategy in that relationship. Such a focus is important since it addresses the blurring of firms' production boundaries and knowledge flows within the supply chain. Future studies might consider investigating the relationship between manufacturing strategy and innovativeness with both outsourcing and subcontracting-in acting as contingent factors in the relationship. Our study did not investigate the indirect relationship between the sourcing strategies and innovation performance. Future studies may investigate the processes through which outsourcing and subcontracting-in relate to innovation performance. For example, it will be interesting to investigate whether factors including absorptive capacity could mediate the relationships between the sourcing strategies and innovation performance. Examples of other factors that merit investigating in tandem with manufacturing strategy include product modularity, industry clock speed and network characteristics. The age of the data might be perceived to be problematic. However, since the relationships investigated in this study are not time dependent and the sourcing strategies tend to be persistent over time, we would not expect the results of our study to be affected by the timing of data collection. Finally, it is worth noting that our dependent variable is a measure of the capability to develop radical innovations (measured by patent count) and not the actual radical innovations developed. Future studies may include measures of actual radical or incremental innovation performance.

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